Processes and Services Programming

Activity 4. Multiprocess Programming Extra

1. Pipelines in Linux: Build a pipeline that lists Python processes sorted by PID.
2. Create a Python program that launches three child processes, each with a different timeout. The parent must wait for all of them to finish.
3. Create a Python program where a producer generates 5 numbers and a consumer processes them using a Queue.
4. Create a Python program where two processes communicate via Pipe. The first sends a message, the second receives it.
5. Create a Python program where two processes modify a shared counter using Value.
6. Create a Python program that calculates the factorial of a large number (e.g. 20) using multiple parallel processes. Divide the calculation into several multiplication ranges. Each process will calculate the partial product of its range. The main process will combine the results to obtain the final factorial.
7. Create a program that divides a list from 1 to 1,000,000 into 4 parts. Each process must calculate the sum of its part (partial sum) and return it. The main process will add the partial results to obtain the total.
8. Create a program that launches three processes that write messages to the same text file (log.txt). Ensure you use a lock to prevent the writes from merging.
9. Simulate a race between 5 processes. Each process advances a random number of steps each turn. The first to reach 100 steps wins.
10. Create a program with two producers that generate numbers and a consumer that processes them. Use a queue.